

Python Code

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1 circle = [(5,1), (2,2), (1,5), (2,8), (5,9), (8,8), (9,5), (8,2)]
2
3 def pattern_circle():
4     print('CIRCLE')
5     for i in range(1, 10):
6         for j in range(1, 10):
7             if (i, j) in circle:
8                 print('*', end=' ')
9             else:
10                print(' ', end=' ')
11        print()
12
13 pattern_circle()
14
15
16 Rhombus = [(1,3), (1,4), (1,5), (1,6), (1,7), (3,2),
17            (3,6), (5,1), (5,2), (5,3), (5,4), (5,5)]
18
19 def pattern_Rhombus():
20     print('RHOMBUS')
21     for i in range(1, 10):
22         for j in range(1, 10):
23             if (i, j) in Rhombus:
24                 print('*', end=' ')
25             else:
26                 print(' ', end=' ')
27        print()
28
29 pattern_Rhombus()
30
31
32 oval = [(3,4), (3,5), (3,6), (4,2), (4,8),
33         (5,1), (5,9), (6,1), (6,9), (7,2), (7,8), (8,4),
34         (8,5), (8,6)]
35
36 def pattern_oval():
37     print('OVAL')
38     for i in range(1, 10):
39         for j in range(1, 10):
40             if (i, j) in oval:
41                 print('*', end=' ')
42             else:
43                 print(' ', end=' ')
44        print()
45
46 pattern_oval()
47
48
49 Triangle = [(2,5), (3,4), (3,6), (4,3), (4,7), (5,2),
50            (5,3), (5,4), (5,5), (5,6), (5,7), (5,8)]
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51
52 def pattern_Triangle():
53     print('TRIANGLE')
54     for i in range(1, 10):
55         for j in range(1, 10):
56             if (i, j) in Triangle:
57                 print('*', end=' ')
58             else:
59                 print(' ', end=' ')
60         print()
61
62 pattern_Triangle()
63
64
65 square = [(2,2), (2,3), (2,4), (2,5), (2,6), (2,7),
66           (2,8), (3,2), (3,8), (4,2), (4,8), (5,2),
67           (5,8), (6,2), (6,8), (7,2), (7,8), (8,2),
68           (8,3), (8,4), (8,5), (8,6), (8,7), (8,8)]
69
70 def pattern_square():
71     print('SQUARE')
72     for i in range(1, 10):
73         for j in range(1, 10):
74             if (i, j) in square:
75                 print('*', end=' ')
76             else:
77                 print(' ', end=' ')
78         print()
79
80 pattern_square()
81
82
83 Semi_circle = [(4,4), (4,5), (4,6), (4,7), (5,3),
84               (5,8), (6,2), (6,9), (7,2), (7,9),
85               (8,2), (8,3), (8,4), (8,5), (8,6),
86               (8,7), (8,8), (8,9)]
87
88 def pattern_SEMI_CIRCLE():
89     print('SEMI CIRCLE')
90     for i in range(1, 10):
91         for j in range(1, 10):
92             if (i, j) in Semi_circle:
93                 print('*', end=' ')
94             else:
95                 print(' ', end=' ')
96         print()
97
98 pattern_SEMI_CIRCLE()
99
100
101 rec = [(1,2), (1,3), (1,4), (1,5), (1,6),
102        (1,7), (1,8), (1,9), (2,2), (2,9),
103        (3,2), (3,9), (4,2), (4,3), (4,4),
104        (4,5), (4,6), (4,7), (4,8), (4,9)]

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105
106 def pattern_rectangle():
107     print('RECTANGLE')
108     for i in range(1, 10):
109         for j in range(1, 10):
110             if (i, j) in rec:
111                 print('*', end=' ')
112             else:
113                 print(' ', end=' ')
114         print()
115
116 pattern_rectangle()
117
118
119 hexagon = [(2,4), (2,5), (2,6), (3,3), (3,7),
120            (4,2), (4,8), (5,1), (5,9), (6,2), (6,8),
121            (7,3), (7,7), (8,4), (8,5), (8,6)]
122
123 def pattern_hex():
124     print('HEXAGON')
125     for i in range(1, 10):
126         for j in range(1, 10):
127             if (i, j) in hexagon:
128                 print('*', end=' ')
129             else:
130                 print(' ', end=' ')
131         print()
132
133 pattern_hex()
134
135
136 trapezoid = [(2,3), (2,4), (2,5), (2,6), (2,7),
137             (3,2), (3,8), (4,1), (4,9), (5,1),
138             (5,2), (5,3), (5,4), (5,5), (5,6),
139             (5,7), (5,8), (5,9)]
140
141 def pattern_trap():
142     print('TRAPEZOID')
143     for i in range(1, 10):
144         for j in range(1, 10):
145             if (i, j) in trapezoid:
146                 print('*', end=' ')
147             else:
148                 print(' ', end=' ')
149         print()
150
151 pattern_trap()
152
153
154 heptagon = [(1,5), (2,3), (2,7), (3,2), (3,8),
155            (4,2), (4,8), (5,2), (5,8), (6,3),
156            (6,7), (7,4), (7,5), (7,6)]
157
158 def pattern_hep():

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159     print('HEPTAGON')
160     for i in range(1, 10):
161         for j in range(1, 10):
162             if (i, j) in heptagon:
163                 print('*', end=' ')
164             else:
165                 print(' ', end=' ')
166         print()
167
168 pattern_hep()
169
170
171 pentagon = [(1,5), (3,3), (3,7), (4,2), (4,8), (6,3), (6,7), (7,4), (7,5), (7,6)]
172
173 def pattern_pen():
174     print('PENTAGON')
175     for i in range(1, 10):
176         for j in range(1, 10):
177             if (i, j) in pentagon:
178                 print('*', end=' ')
179             else:
180                 print(' ', end=' ')
181         print()
182
183 pattern_pen()
184
185
186 octagon = [(2,4), (2,5), (2,6), (3,3), (3,7), (4,2), (4,8), (5,2), (5,8), (6,2),
187            (6,8), (7,3), (7,7), (8,4), (8,5), (8,6)]
188
189 def pattern_oct():
190     print('OCTAGON')
191     for i in range(1, 10):
192         for j in range(1, 10):
193             if (i, j) in octagon:
194                 print('*', end=' ')
195             else:
196                 print(' ', end=' ')
197         print()
198
199 pattern_oct()
200
201
202 heart = [(2,2), (2,3), (2,7), (2,8), (3,1), (3,4), (3,6), (3,9), (4,1), (4,9), (5,2),
203          (5,8), (6,3), (6,7), (7,4), (7,6), (8,5)]
204
205 def pattern_heart():
206     print('HEART')
207     for i in range(1, 10):
208         for j in range(1, 10):
209             if (i, j) in heart:
210                 print('*', end=' ')
211             else:
212                 print(' ', end=' ')

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211         print()
212
213     pattern_heart()
214
215
216     nonagon = [(1,5), (2,3), (2,7), (4,1), (4,9), (6,1), (6,9), (8,3), (8,7), (9,4),
217                (9,5), (9,6)]
218
219     def pattern_non():
220         print('NONAGON')
221         for i in range(1, 10):
222             for j in range(1, 10):
223                 if (i, j) in nonagon:
224                     print('*', end=' ')
225                 else:
226                     print(' ', end=' ')
227             print()
228
229     pattern_non()
230
231     decagon = [(1,4), (1,5), (1,6), (2,3), (2,7), (3,2), (3,8), (4,2), (4,8), (5,2),
232               (5,8), (6,2), (6,8), (7,3), (7,7), (8,4), (8,5), (8,6)]
233
234     def pattern_dec():
235         print('DECAGON')
236         for i in range(1, 10):
237             for j in range(1, 10):
238                 if (i, j) in decagon:
239                     print('*', end=' ')
240                 else:
241                     print(' ', end=' ')
242             print()
243
244     pattern_dec()
245
246     star = [(1,5), (2,3), (2,6), (3,2), (3,8), (4,3), (4,7), (5,2), (5,5), (5,8), (6,1),
247            (6,3), (6,7), (6,9)]
248
249     def pattern_star():
250         print('STAR')
251         for i in range(1, 10):
252             for j in range(1, 10):
253                 if (i, j) in star:
254                     print('*', end=' ')
255                 else:
256                     print(' ', end=' ')
257             print()
258
259     pattern_star()
260
261     smiley = [(3,3), (3,7), (6,2), (7,3), (8,4), (8,5), (8,6), (7,7), (6,8)]

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262
263 def pattern_smiley():
264     print('SMILEY FACE')
265     for i in range(1, 11):
266         for j in range(1, 11):
267             if (i, j) in smiley:
268                 print('+', end=' ')
269             else:
270
271                 if (i==1 and j>3 and j<8) or (i==10 and j>3 and j<8) or \
272                     (j==1 and i>3 and i<8) or (j==10 and i>3 and i<8) or \
273                     (i==2 and (j==2 or j==9)) or (i==9 and (j==2 or j==9)):
274                     print('.', end=' ')
275                 else:
276                     print(' ', end=' ')
277         print()
278
279 pattern_smiley()
280
281
282
283 heart_outer = [
284     (2,4), (2,5), (2,8), (2,9),
285     (3,3), (3,6), (3,7), (3,10),
286     (4,2), (4,11),
287     (5,2), (5,11),
288     (6,3), (6,10),
289     (7,4), (7,9),
290     (8,5), (8,8),
291     (9,6), (9,7)
292 ]
293
294 heart_inner = [
295     (3,4), (3,5), (3,8), (3,9),
296     (4,3), (4,6), (4,7), (4,10),
297     (5,3), (5,10),
298     (6,4), (6,9),
299     (7,5), (7,8),
300     (8,6), (8,7)
301 ]
302
303 def pattern_double_heart():
304     print('\nDOUBLE LAYER HEART')
305     for i in range(1, 13):
306         for j in range(1, 13):
307
308             if (i, j) in heart_inner:
309                 print('*', end=' ')
310             elif (i, j) in heart_outer:
311                 print('+', end=' ')
312             else:
313                 print(' ', end=' ')
314         print()
315

```

316 pattern_double_heart()

>>>

===== RESTART: C:/Python314/GEN-AI-VL-A6.3.py =====

CIRCLE

```
      *
    *      *
*           *
      *      *
    *      *
```

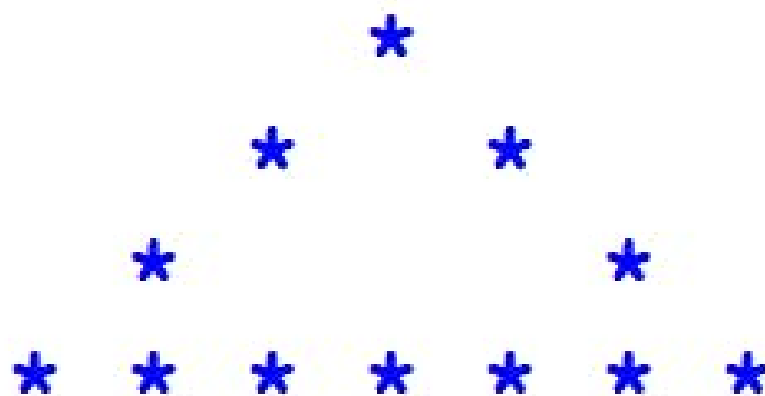
RHOMBUS

```
    * * * * *
  *           *
* * * * *
* * * * *
```

OVAL



TRIANGLE



SQUARE



SEMI CIRCLE



RECTANGLE



HEXAGON



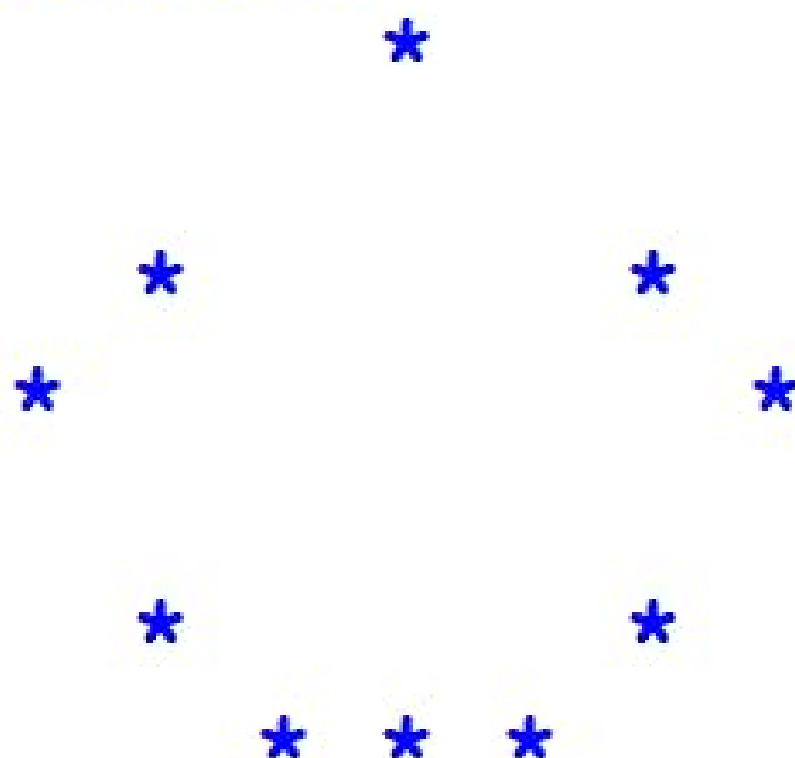
TRAPEZOID



HEPTAGON



PENTAGON



OCTAGON



HEART



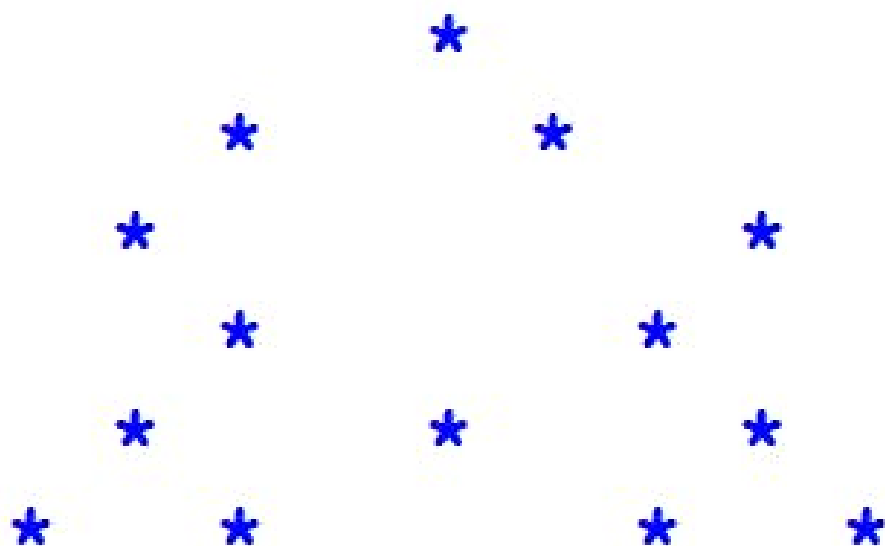
NONAGON



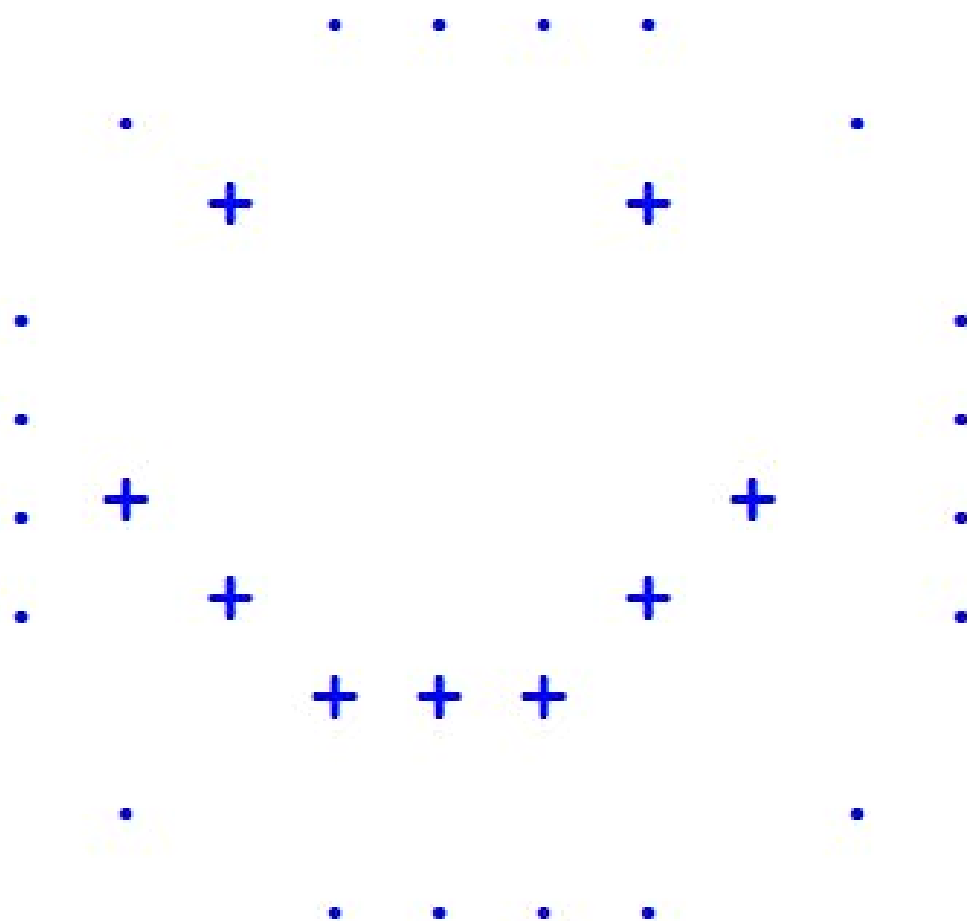
DECAGON



STAR



SMILEY FACE



DOUBLE LAYER HEART

